Signed horizons and residue-encoded Pell witnesses

Explicit Diophantine equations for the consistency of ZFC

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Abstract

We give new explicitly instantiated Diophantine equations whose unsolvability, in their stated domains, is provably equivalent in ZFC to its syntactic consistency. A signed affine proof horizon removes an integer variable from each of two directly compiled quartics. The balanced result has 214 unrestricted integer variables, 153 squared quadratic residuals, 5,875 collected monomials and coefficient height 655,360. A second quartic has 212 integer variables and larger coefficients. An independent Pell-ordinate construction replaces a quantified quotient by a residue already carried by an auxiliary ordinate. It yields fourteen positive variables at 66 operations and degree 174, or 68 operations and degree 146. A careful domain audit turns the latter into a fully explicit equation in 36 unrestricted integers, degree 288 and 145 operations. Composing signed horizons with a reused CRT anchor and oriented power pooling also gives eight direct integer profiles, including 117 variables at degree 16, 108 at degree 32, and 86 at degree 2,097,780. All final equations have fixed coefficients and contain only polynomial operations. The new proofs are ordinary mathematics formalizable in ZFC, composed with explicitly identified inherited calibration theorems. Exact finite verification is supplied, but no new end-to-end Lean/Rocq or encoded object-language ZFC proof is claimed.

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1 Results, interpretation, and comparison scope

The task is to make the consistency equation simpler while keeping both directions of its calibration. There is no single order on simplicity: variable count, total degree, evaluation length, coefficient height, coefficient storage and proof complexity can move in opposite directions. This submission therefore gives exact completed constructions and makes the relevant tradeoffs explicit. It does not prove global optimality or claim worldwide priority for its component techniques.

The repository comparison is pinned to

`02a78a08603cbae248c19c23a75907c6c6bd6443`.

The immediate direct source is package 173 [1]; the compact source is package 174 [2], with a residue-carrier argument from package 170 [3]. The direct higher-degree family also uses the separately delivered *Oriented Powers* packet [4]. Its archive identity is retained in the source ledgers.

1.1 The precise logical claim

Fix the effective first-order presentation denoted `ZFCax_s` and the finite proof relation `BProv` in the repository. Let $\mathrm{Ref}_{\mathrm{ZFC}}$ mean that there is a finite derivation of `Form.fBot` from that presentation, and put $\mathrm{Con}(\mathrm{ZFC}) = \neg\,\mathrm{Ref}_{\mathrm{ZFC}}$. The new constructions prove the following ordinary theorem.

Theorem 1.1. *For every final polynomial P delivered with this article and its specified domain $\mathcal{D}$, ZFC proves*

$$\big(\exists \mathbf{x} \in \mathcal{D}\ P(\mathbf{x}) = 0\big) \longleftrightarrow \mathrm{Ref}_{\mathrm{ZFC}}\,.$$

Consequently ZFC proves

$$\boxed{\big(\neg\exists \mathbf{x} \in \mathcal{D}\ P(\mathbf{x}) = 0\big) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).}$$

The coefficients are fixed independently of the truth of consistency.

The theorem proves an equivalence. It does not establish the absence of zeros, nor consistency itself. Its quantifiers are the internally coded integers and finite derivations of ZFC. The argument is not restricted to externally standard proof lengths.

1.2 The useful completed endpoints

Equation family	Domain	Variables	Exact degree	Binary operations
Balanced direct quartic Q_{214}	$\mathbb{Z}$	214	4	—
Dimension-focused quartic Q_{212}	$\mathbb{Z}$	212	4	—
Compact P_E	$\mathbb{Z}_{>0}$	14	174	66
Compact P_q	$\mathbb{Z}_{>0}$	14	158	67
Compact P_q°	$\mathbb{Z}_{>0}$	14	146	68
Explicit compact integer lift P_{36}	$\mathbb{Z}$	36	288	145
Direct pooled profile	$\mathbb{Z}$	132	12	2484
Direct pooled profile	$\mathbb{Z}$	117	16	2439
Direct pooled profile	$\mathbb{Z}$	108	32	2434
Smallest delivered direct profile	$\mathbb{Z}$	86	2097780	2385

Operation counts are binary $+, -, \times$ gates in the supplied straight-line programs, with fixed coefficients as input numerals and shared intermediate values counted once. They measure arithmetic circuit length; big-integer bit complexity is a separate cost.

The compact integer equation P_{36} has a short factorized definition: Section 3 gives all its polynomial factors and Section A defines its four fixed constants. The balanced quartic has modest *expanded*

coefficients and no native coefficient ports; Appendix B prints every residual. The direct circuits are useful alternatives when avoiding the large compact constants matters more than arithmetic evaluation length.

The fourteen-positive-variable equations do not assert a fourteen-*integer*-variable theorem. The stated conversion to 36 integer variables is part of the proof. Likewise, the 86-variable direct equation is not a variable-count record across other encoding families. Much smaller-variable equations at other degree and coefficient costs already appear in the repository. Published small-variable universality results also have a different optimization objective [8].

1.3 How this iteration changes the earlier results

The balanced parent from package 173 has 215 integer variables, 154 quadratic residuals, 5,660 monomials and height 91,679. The new balanced quartic removes one variable and one residual, at the cost of increasing support to 5,875 and height to 655,360. The 213-variable companion similarly becomes 212 variables, with increased coefficient height. These are dimension reductions, not improvements in every metric.

The compact parent has fifteen positive variables, degree 157 and 66 operations. The new P_E keeps 66 operations with fourteen variables, while its degree is 174. The lower-degree P_q° improves the documented fourteen-positive 71-operation, degree-157 comparison to 68 operations and degree 146. Complete coefficient tuples are kept separate throughout this comparison.

The pooled direct family has stronger matched improvements: 137 variables at degree 12 become 132 with the same bound of 256 on literal circuit constants; 121 at degree 16 become 117; 111 at degree 32 become 108. Its final 86-variable profile has degree 2,097,780, compared with 87 variables and degree 8,388,612 in the earlier packet. The expanded coefficient bounds remain a separate column of the exact machine-readable comparison.

2 A signed horizon removes a quartic variable

The old direct source imposed a positive quadratic horizon such as $t = 16 + 4z^2$ or $t = 128 + 4z^2$. This guaranteed positivity at the cost of a coordinate that could not be removed affinely. The power machinery already contains enough information to recover the sign. We use that information and replace the square progression by a signed affine progression.

2.1 A uniform signed-horizon lemma

Put $\ell = 2^{20}$. Fix an integer $m \geq 8$ divisible by four and impose

$$t = m(2z + 1), \qquad z \in \mathbb{Z}.$$

Then $t \neq 0$, $|t| \geq m$ and $4 \mid t$. Retain the source's signed Pell frame, its reused even CRT index C, all orientation rows and all actual power requests. Two growth choices will be used:

$$K = \ell t, \tag{1}$$

$$U = at^2, \qquad K = bU^2 \quad (a, b \in \mathbb{Z}_{>0},\ ba^2m^3 \geq \ell). \tag{2}$$

The retained Pell request expresses $F = t^K$. This exponentiation is part of the *proof interpretation*, not the final polynomial.

Lemma 2.1. *At every integer zero of either modified source, $t \geq m$ and $K \geq \ell t$. Every finite C222 refutation can be represented using the new positive branch of the horizon progression.*

Proof. The source's preliminary signed anchor argument uses squares and absolute values. It recovers the canonical signed Pell pairs before decoding arithmetic signs. Its actual frame includes the squares of all requested bases, outputs and indices, in particular the changed index K. Thus it still gives

$$W \geq 258, \qquad |A| > W^W.$$

For the affine index this lower bound on W already follows from $K^2 \geq \ell^2 m^2$; for the quartic index it follows from $K = ba^2t^4$. The signed power lemma therefore applies at a nonzero base. At a base of absolute value at least two, an integral exact signed-index power must have nonnegative exponent.

For (1), $|t| \geq 8$ makes the first power force $K \geq 0$. Since $K = \ell t$ and $t \neq 0$, this proves $t \geq m$ immediately.

For (2), the index $K = ba^2t^4$ is positive and even before the sign of t is known. The first power gives $F = t^K > 0$, with F even and $F > |t|$. The retained CRT row is

$$Bj = C + F + 1, \qquad C = 2c.$$

Its right side is odd, so B is odd and nonzero. The self-radix multiplier has the form $H = 2 + 4S$ for an integer S, hence $|BH| \geq 2$. The exact power at that base gives $x \geq 0$ and $q = (BH)^x$.

It remains to rule out $x = 0$. The beta endpoint rows, which were never deleted, give

$$y = F(2v + 1), \qquad g = 1 + (t + 1)y, \qquad x = 1 + gu_t.$$

Because $|y| \geq F$ and $|t + 1| \geq m - 1 \geq 7$,

$$|g| \geq (m - 1)F - 1 > 1.$$

Thus $x = 0$ would imply $gu_t = -1$, impossible in integers. We have $x \geq 1$ and consequently $|q| \geq 2$. The literal retained request is q^t. Its signed exactness forces $t \geq 0$, and the progression then gives $t \geq m$. Notice that the proof does not replace the digit row by a statement about B alone; its unrestricted high quotient would invalidate that shortcut.

For either growth choice, positivity now yields $K \geq \ell t$. Completeness uses the cofinal positive progression $m(2z + 1)$: choose t above the finite proof length and the local free parameters, then use the source construction with this t. The numerical estimates and the order of the subsequent choices are justified below. □

2.2 The estimates needed at minimum eight

The literal square-shell constructors obey a sharper bound than the earlier conservative $t \geq 16$ presentation. For integers $0 \leq r, s < T$,

$$E(r, s) \leq 8T^2 - 2T + 1 < 8T^2,$$

and the other three constructors are no larger. Thus, if both arguments are strictly below t^e and $t \geq 8$, every constructor output is strictly below t^{2e+1}. With $e_0 = 1$ and $e_{d+1} = 2e_d + 1$, one obtains $e_d = 2^{d+1} - 1$.

The authenticated source constructor depths are

$$4,\ 3,\ 3,\ 3,\ 3,\ 5,\ 4,\ 3,\ 19.$$

Consequently every local formula label is below $t^{2^{20}-1}$. Set $L = t^\ell$. Then $L \geq t$ and $F = t^K \geq L^t$. No bound on the deleted sum of derived labels is needed here.

Lemma 2.2 (Boundary-valid label separation)**.** *Let t, L, F be positive integers with $L \geq t$ and $F \geq L^t$. If two multisets of labels in $[1, L]$, with at most t and $2t + 1$ entries respectively, satisfy*

$$\prod_{j \in U} (1 + jF) = \prod_{j \in V} (1 + jF),$$

then their label multiplicities agree.

Proof. If the factor counts differ, let r be the smaller count. Then $r \le t$, and comparison of the two products gives

$$(F+1)^{r+1} \le (1+LF)^r \le L^r(F+1)^r,$$

contradicting $F+1 > L^t$. Their common count s is therefore at most t. If $s=0$, both multisets are empty; assume henceforth $s \ge 1$. Each coefficient of either product polynomial in an indeterminate lies between 1 and

$$\binom{s}{k} L^k \le s^{s-k} L^k \le L^s \le F.$$

Thus every coefficient of their difference has absolute value strictly below F, even when one coefficient equals F. The least nonzero coefficient of a difference vanishing at F would have to be divisible by F, a contradiction. The product polynomials agree. Their roots $-1/j$ recover the multiplicities. □

Truth has label zero and hence weight $1 + 0F = 1$; its factors are omitted before applying the positive-label lemma. The source's multiplicative Horn updates therefore recover the same finite rule-flow balance. If falsity were outside the closure of truth under the selected rules, every outside conclusion would require an outside premise. Summing that balance over outside labels would be nonpositive, whereas the target falsity contributes one. This yields a finite C222 refutation even if the selected rules have cycles.

Two other conservative numerical estimates have explicit replacements. Since $K \ge \ell t$ and $t \ge 8$,

$$F \ge 8^{8\ell} > 2^{64}.$$

After the unchanged beta sign recovery, $y \ge F$ and $x \ge 2 + (t+1)y$. Therefore $2 + t + x + y + F < 2x$, $x > 2^{32}$ and the retained local polynomial budget gives

$$\text{Local} < (2x)^{2^{23}} < 2^x.$$

Also

$$F \ge 8^{2t} > 16^t > \operatorname{lcm}(1, \dots, t)^2.$$

This supplies the radius for the inherited factorial-free binomial CRT selector. No factorial-rounding lemma is being applied outside its range.

Finally $v_2(F) = K v_2(t) \ge \ell t$ exceeds both $\lfloor \log_2 t \rfloor$ and $v_2(t!)$. Hence $\binom{F+t}{t}$ is odd. Multiplying F by a suitable positive odd multiple supplies a beta spacing divisible by $t!$ and larger than all finite pending states. After choosing the beta code and the every-prime CRT residues, choose the encoded index C even and larger than all other requested indices. Only then choose H, q and the Pell frame. The choices are cofinal and noncircular. These facts complete the soundness and completeness assertions in Lemma 2.1.

2.3 Two literal quartics

The dimension-focused choice is

$$t = 8 + 16z, \qquad K = 1048576t.$$

Affine elimination of t gives Q_{212} with 212 integer coordinates, 151 quadratic residuals and 5,706 collected monomials. Its exact expanded coefficient height is

$$3169126500588047470295983718 40.$$

The balanced choice is

$$t = 8 + 16z, \qquad U = 2t^2, \qquad K = 512U^2 = 2048t^4.$$

At $t = 8$ the inequality $2048t^3 \geq 2^{20}$ is already an equality. This gives Q_{214} with 214 integer coordinates, 153 residuals, 5,875 collected monomials and exact coefficient height 655,360. Its maximum residual coefficient is 512.

For each profile, replace the source's horizon row by $t - m - 2mz = 0$, substitute $t = m + 2mz$ into every other row, and delete this row and the coordinate t. This is a polynomial bijection on zeros of the *modified* source, with inverse restoring the displayed t. Affine substitution keeps every residual quadratic.

For the balanced profile, change only the two protected growth rows to $U - 2t^2$ and $K - 512U^2$ before eliminating t. Subtract twice the square-definition row from the frame row, and add the square-definition row to the orientation row. These integral row operations have explicit inverses and remove duplicated t^2 terms.

The finite checker also verifies that these edits commute with the inherited six coordinate eliminations and optimization chain. The radical directions vanish on the protected coordinates; the affine projection uses only the stated local constructor rows; later charts leave the horizon and growth coordinates unchanged; and row operations do not modify their defining rows. Thus the emitted equations implement the stated arithmetic source change, rather than an unsupported edit to an optimized polynomial.

Corollary 2.3. *For $r = 212, 214$, ZFC proves*

$$\neg\exists \mathbf{x} \in \mathbb{Z}^r \; Q_r(\mathbf{x}) = 0 \quad \longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

The complete Q_{214} residual list appears in Appendix B; both collected polynomials are in `quartic/`.

3 An auxiliary ordinate removes a quantified quotient

The next construction improves the inexpensive positive-variable branch. It combines the normalized auxiliary ordinate of the fifteen-variable P66 equation [2] with the residue-carrier argument developed for combined Pell roots in package 170 [3]. The resulting equations have fourteen positive unknowns and the following exact profiles. The whole Start-only4127 coefficient tuple is retained unchanged; its complete finite definition is in Section A.

Equation	Positive unknowns	Binary operations	Degree
Immediate P66 parent	15	66	157
Previously documented comparison	14	70	189
Previously documented comparison	14	71	157
New P_E	14	66	174
New P_q	14	67	158
New P_q°	14	68	146

The first new row reduces the dimension of the immediate parent at the same arithmetic cost, with a degree increase. The first and last new rows also improve both cost and degree against the corresponding fourteen-variable comparisons. The previous comparison tuples and the new tuple are distinct complete coefficient specializations. These are documented improvements within this branch, not a global minimum theorem.

3.1 Complete polynomial definitions

The independent positive integers are

$$(J, F, \alpha, z, f, n, i, T, s, w, \tau, \eta, \zeta, \gamma).$$

With the four fixed integers β, K_0, M_C, M_F, define

$$\begin{aligned} q &= \beta J + 1, & C &= q - F - \alpha, & X &= wq, & Y &= sq^3, \\ E &= XY, & k &= \eta + \zeta, & \ell &= EkY, & c &= kY + \eta, \\ a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1, \end{aligned}$$

and

$$\begin{aligned} D &= X + ac + \gamma H, \\ R &= [(q-1)(q-F) + \alpha](q^2-1) + (M_C + qM_F)J, \\ V &= f^2(cT - R) - c, \\ N_0 &= \tau^2 - \ell(\ell + k), \qquad N_1 = D^2 - \Delta c^2, \\ N_t &= (K_0 + w)C + (q - F) - z(q-1). \end{aligned}$$

Choose either $(M, d) = (E, E)$ or $(M, d) = (q^4, q^2)$ and put

$$\begin{aligned} u &= k - Mn - R, & I &= dic^2, & S &= \Delta I, \\ N_a &= S^2(V^2 - u^2) + u^2, \\ N_s^{\mathrm{sc}} &= \Delta f^2 - S^2, & N_s &= f^2 - \Delta I^2. \end{aligned}$$

The displayed letters are polynomial abbreviations, not additional unknowns. For $(M, d) = (E, E)$ the equation is

$$\boxed{P_E = N_0 N_1 N_a N_t N_s^{\mathrm{sc}} - \Delta = 0.} \tag{3}$$

For $(M, d) = (q^4, q^2)$ we supply both

$$P_q = N_0 N_1 N_a N_t N_s^{\mathrm{sc}} - \Delta = 0. \tag{4}$$

$$\boxed{P_q^\circ = N_0 N_1 N_a N_t N_s - 1 = 0.} \tag{5}$$

The exact identity $P_q = \Delta P_q^\circ$ and $\Delta > 0$ show that these last two equations have the same positive zero set. The unscaled strong norm costs one additional multiplication but reduces degree by twelve.

The equivalence theorem. For each of these fixed polynomials, ZFC proves that a positive zero exists if and only if a finite syntactic ZFC refutation exists. Consequently ZFC proves that the stated positive zero set is empty if and only if Con(ZFC). The proof below restores a literal parent P66 zero and constructs new witnesses from its canonical completeness data. The native and foundational calibrations are the identified inherited theorems, not conclusions of a finite numerical search.

3.2 Unit signs are resolved before recovering an index

At a positive zero, canceling Δ in the proof, if necessary, gives

$$N_0 N_1 N_a N_t N_s = 1. \tag{6}$$

Every factor is an integer unit. Let $v = XY^2 = EY$. Then $N_0 = \tau^2 - v(v+1)k^2$. For $v \geq 2$, the negative Pell equation $x^2 - v(v+1)y^2 = -1$ is impossible: a solution with least positive y satisfies $vy < x < (v+1)y$, and multiplication by $(2v+1) - 2\sqrt{v(v+1)}$ produces another solution with ordinate $0 < |(2v+1)y - 2x| < y$. Thus $N_0 = 1$.

The number $\Delta = (a+2)^2 - 1$ is 0 or 3 modulo 4, so neither N_1 nor N_s can be -1. Modulo 4, N_a is congruent to u^2 or V^2, according as S is even or odd, so it cannot be -1 either. Equation (6) therefore yields

$$N_0 = N_1 = N_a = N_t = N_s = 1. \tag{7}$$

In particular, deleting the old quotient factor removes its previously unresolved partner sign in N_t. We immediately have

$$(K_0 + w)C = 1 + z(q-1) - (q-F) \geq F > 0,$$

so $C > 0$. The unchanged coherent coefficient tuple satisfies $0 < M_C < \beta < M_F < 2\beta$. Applying these inequalities to the literal definition of R, with $G_0 = q\alpha + (q-1)C$, gives

$$3q < R < q^4 - q^3 + q^2 - 1 < q^4 - 2 \leq E - 2. \tag{8}$$

In both variants $R < M$. None of these statements has used the omitted quotient or assumed a main Pell index.

3.3 The main index is authenticated without that quotient

For $b \geq 2$, write $\chi_b(j), \psi_b(j)$ for the Pell sequences with initial values $(1, b)$ and $(0, 1)$ and recurrence $U_{j+2} = 2bU_{j+1} - U_j$. Put $A = a + 2$ and $P_0 = 1 + 2v$. The norm equations imply

$$\tau = \chi_{P_0}(r), \qquad k = 2\psi_{P_0}(r), \qquad r \geq 1, \tag{9}$$

$$D = \chi_A(p), \qquad c = \psi_A(p), \qquad p \geq 1. \tag{10}$$

For the first norm the fundamental positive solution is $(2v+1, 2)$; ordinate one would put a square strictly between v^2 and $(v+1)^2$. The case $r = 1$ would give $k = 2$, $\eta = \zeta = 1$, and $1 < c = 2Y + 1 < 2A$, which is impossible in (10). Hence

$$r \geq 2, \quad k \geq 8XY^2 + 4, \quad c > kY > 8XY^3 > 2R. \tag{11}$$

Since $P_0 > A$ and $c > k$, monotonicity gives $r < p$; now $p \geq 3$ and $A \geq 2$ imply $c = \psi_A(p) > 2p$.

The strong equation gives $f = \chi_A(m)$ and $I = \psi_A(m)$. The literal divisibility $c^2 \mid I$ and strong divisibility first imply $p \mid m$. Writing $m = pj$, composition gives $I = c\psi_D(j)$. Since $D^2 \equiv 1 \pmod{c}$, the recurrence gives $\psi_D(j) \equiv jD^{j-1} \pmod{c}$. Thus $c \mid j$ and

$$pc \mid m, \qquad c \mid m, \qquad m > 2p. \tag{12}$$

Let $Q = S^2 = \Delta(f^2 - 1) > 1$. The auxiliary equation is

$$QV^2 - (Q-1)u^2 = 1.$$

Neither coordinate is zero; also $V > 0$ by (8)–(11), $T \geq 1$, and $f^2 > 2$. Therefore $SV = \chi_S(\nu)$ and $|u| = \psi_S(\nu)$ for a positive odd index ν. An even index would make the abscissa ± 1 modulo S.

For $\nu = 2j+1$, the integer polynomials $L_0(Z) = 1$, $L_1(Z) = 4Z - 3$, $L_{j+1}(Z) = (4Z-2)L_j(Z) - L_{j-1}(Z)$ satisfy

$$\chi_b(2j+1) = bL_j(b^2), \qquad L_j(0) = (-1)^j(2j+1),$$

$$L_j(1 - A^2) = (-1)^j \psi_A(2j+1).$$

Using $Q \equiv 1 - A^2 \pmod{f}$ and $V \equiv -c \pmod{f}$ yields $\psi_A(\nu) \equiv \pm c \pmod{f}$. Squaring gives

$$\chi_A(2\nu) \equiv \chi_A(2p) \pmod{\chi_A(m)}.$$

The signed step-down argument from [3] applies with $0 < 2p < m$: addition identities reduce any abscissa to a signed $\chi_A(t)$ with $0 \le t \le m$; the endpoint has residue zero, $2\chi_A(m-1) < \chi_A(m)$ excludes a negative representative, and monotonicity identifies the positive one. It follows that $2\nu \equiv \pm 2p \pmod{2m}$ and hence $\nu \equiv \pm p \pmod{c}$. Separately, $Q \equiv 0 \pmod{c}$ and $V \equiv -R \pmod{c}$ give $\nu \equiv \pm R \pmod{c}$. As $0 < p, R < c/2$, we conclude

$$p = R. \tag{13}$$

The omitted quotient has not occurred in any premise of this authentication.

3.4 The ordinate restores the missing quotient

Both choices satisfy $M \mid v$ and $M \mid S^2$. Reducing the first recurrence modulo M gives $k \equiv 2r \pmod{M}$. The odd-ordinate identity

$$\psi_S(2j+1) \equiv (-1)^j \pmod{4S^2} \tag{14}$$

follows by induction. Thus $u \equiv \pm 1 \pmod{M}$. Since $u = k - Mn - R$, there is $\varepsilon \in \{1, -1\}$ such that $2r - R \equiv \varepsilon \pmod{M}$. Now $r < p = R < M$ implies $|2r - R - \varepsilon| < R < M$, so

$$2r = R + \varepsilon.$$

The negative choice would give $R = 2r + 1$. But $B_0 = 2A^2 - 1 > P_0$ and composition then imply

$$c = \psi_A(2r+1) > 2A\psi_{B_0}(r) \ge 2A\psi_{P_0}(r) = Ak > k(Y+1),$$

contrary to $c = kY + \eta < k(Y+1)$. Therefore $2r = R+1$. The full first recurrence gives $k \equiv 2r \pmod{4v}$, so

$$h_{\text{old}} = \frac{k - R - 1}{E} \in \mathbb{Z}_{>0}. \tag{15}$$

Positivity follows from (11) and (8). Together with $i_{\text{old}} = di$ and $y_{\text{old}} = |u| > 0$, this restores an exact zero of the original P66 equation: its omitted quotient factor is $+1$, and all its other normalized factors are $+1$. The retained whole-tuple calibration supplies a finite syntactic refutation.

For completeness, start with a refutation and its canonical outer P66 witnesses. The inherited native theorem supplies $R \equiv 3 \pmod 4$, $c = \psi_A(R)$, $k = 2\psi_{P_0}((R+1)/2)$, and the required outer unit factors. Choose a positive period of the determinant-one Pell matrix modulo $c^2 d$. It supplies $m > 0$ such that $c^2 d \mid \psi_A(m)$. Set $f = \chi_A(m)$ and $i = \psi_A(m)/(c^2 d) > 0$ before choosing fresh auxiliary witnesses. Define

$$V_* = \frac{\chi_S(R)}{S}, \qquad y_* = \psi_S(R).$$

These are positive integers. The same polynomial identities, now modulo f^2 and c, give $V_* \equiv -c \pmod{f^2}$ and $V_* \equiv -R \pmod{c}$. Since $\gcd(c, f) = 1$ and $f^2 \equiv 1 \pmod{c}$,

$$T = \frac{V_* + c + Rf^2}{cf^2} \in \mathbb{Z}_{>0}.$$

Finally (14) gives $y_* \equiv -1 \pmod{M}$, whereas $k - R \equiv 1 \pmod{M}$. Thus

$$n = \frac{y_* + k - R}{M} \in \mathbb{Z}_{>0}, \qquad u = -y_*.$$

Every new coordinate is positive and every unit factor is $+1$. This proves completeness with newly chosen private witnesses, rather than assuming old private witnesses retain stronger divisibilities.

3.5 Exact costs and nonzero leading forms

In P_E the parent's three gates for $k - hE - R$ are reused as $k - nE - R$. Removing this factor from the final product saves one multiplication; scaling S by E costs one. Thus the count remains 66. The q^4 modulus adds one gate, giving 67; computing SI for the unscaled strong norm adds one further gate, giving 68. All inputs and gates are live.

Write $q_0 = \beta J$, $k_0 = \eta + \zeta$ and $B_0 = w(q_0 - F - \alpha) - zq_0$. The complete leading homogeneous forms are

$$\begin{aligned}
[P_E]_{174} &= 8q_0^{103}\gamma i^4 k_0^{13} w^{16} s^{29} T^2 f^4 B_0,\\
[P_q]_{158} &= 8q_0^{95}\gamma i^4 k_0^{13} w^{12} s^{25} T^2 f^4 B_0,\\
[P_q^\circ]_{146} &= 8q_0^{87}\gamma i^4 k_0^{13} w^{10} s^{23} T^2 f^4 B_0.
\end{aligned}$$

The only leading cancellation is accounted for exactly by

$$N_1 = 2acZ + Z^2 - Hc^2, \qquad Z = X + \gamma H,$$

which has degree 18. The five factor degrees are respectively $(22, 18, 74, 2, 58)$, $(22, 18, 66, 2, 50)$ and $(22, 18, 66, 2, 38)$. Choosing the $-zq_0$ term and the η^{13} term displays coefficients $-8\beta^{104}$, $-8\beta^{96}$ and $-8\beta^{88}$. They are nonzero at the actual fixed β, certifying the claimed exact degrees.

3.6 A complete equation in 36 unrestricted integers

The sign audit permits three parent coordinates to range over all integers. The polynomial uses f and τ only squared, and their recovered norms force them to be nonzero, so absolute values restore positivity. The sign of n was never used in soundness: only the signed carrier u and its residue modulo M matter. Completeness already constructs these three coordinates positively. Thus only

$$J, F, \alpha, z, i, T, s, w, \eta, \zeta, \gamma$$

require positivity encodings.

For each coordinate x in this eleven-element list, substitute

$$x = 1 + x_a^2 + x_b^2 + x_c(x_c + 1), \qquad x_a, x_b, x_c \in \mathbb{Z}, \tag{16}$$

in P_q° and leave f, n, τ unrestricted. This is the explicit polynomial P_{36} in the supplied literal graph. The parameterization is positive for every integer triple and represents every positive integer. For a short derivation, Legendre's three-square theorem applies to $4N + 1$ for every $N \geq 0$; reducing modulo 4 shows that exactly one square is odd, so after permutation

$$4N + 1 = (2a)^2 + (2b)^2 + (2c + 1)^2, \quad N = a^2 + b^2 + c(c + 1).$$

This representation is also explicitly formalized in the Archive of Formal Proofs [7]. We obtain the ordinary ZFC theorem

$$\boxed{\mathrm{ZFC} \vdash (\neg\exists \mathbf{x} \in \mathbb{Z}^{36}\ P_{36}(\mathbf{x}) = 0) \iff \mathrm{Con(ZFC)}.} \tag{17}$$

No inequalities or extra quantified registers remain in P_{36}.

There are $3 \cdot 11 + 3 = 36$ integer unknowns. Each substitution costs seven binary operations, giving $68 + 7 \cdot 11 = 145$ operations. Giving the eleven substituted coordinates weight two and f, n, τ weight one yields factor degrees $(44, 36, 128, 4, 76)$, totaling 288. The squared carrier contributes only degree 118 in the auxiliary norm, so it cannot overtake the degree-128 V^2 term. The complete leading form is the last

displayed positive leader with each substituted x replaced by $\sigma_x = x_a^2 + x_b^2 + x_c^2$ and with f unchanged. Setting all x_b, x_c to zero exhibits the nonzero coefficient $-8\beta^{88}$ at

$$J_a^{176}\gamma_a^2 i_a^8 \eta_a^{26} w_a^{20} s_a^{46} T_a^4 f^4 z_a^2.$$

Its exponents sum to 288, so this degree is exact.

The 36-integer equation uses a large fixed coefficient architecture. Its size should therefore be compared separately with small-coefficient direct quartics and with still smaller but higher-degree constructions. The new finite checks authenticate the literal formulas, full fixed bytes, liveness, and leading forms; the unbounded equivalence is the mathematical proof above composed with the retained native/refutation calibration.

4 Composing signed horizons with oriented power pools

The reused CRT anchor and the oriented power theorem have complementary strengths. The former removes a derived-label range request; the latter combines the remaining power certificates. Their composition requires a new sign argument, because the factorial normalization used in the earlier pooling proof is absent after the anchor reduction.

4.1 The common arithmetic interface

The five bases are

$$n_0 = t, \qquad n_1 = B, \qquad n_2 = BH, \qquad n_3 = q, \qquad n_4 = q + 1,$$

with respectively $1, 1, 1, 3, 10$ power requests. The retained canonical signed Pell data at center A satisfy

$$|A| > W^W, \qquad n_j^2, m_i^2, k_i^2 < W, \qquad W \geq 258.$$

This interface is established before the signs of the arithmetic coordinates are decoded. Every requested index has a canonical signed Pell pair (X_i, Y_i), with $D_i = X_i - AY_i$.

Define, for each block,

$$\begin{aligned} T_j &= 2An_j - n_j^2 - 1, \\ E_j &= \sum_{i \in j} ((m_i - D_i)^2 - Y_i^2) - 2n_j \sum_{i \in j} Y_i(m_i - X_i). \end{aligned}$$

The five-base residual is the ordinary polynomial

$$R = \sum_{j=0}^{4} n_j E_j \prod_{\ell \neq j} T_\ell - z \prod_{j=0}^{4} T_j.$$

The inherited oriented-pooling theorem makes $R = 0$ equivalent to exactness of every request at a nonzero actual base. Zero-base blocks are temporarily unconstrained, and are excluded by the argument below. For nonzero bases, $\operatorname{sgn}(n_j/T_j) = \operatorname{sgn}(A)$; this common sign is what prevents cancellation of the squared power errors.

Two smaller-degree alternatives use three and two quotient coordinates. The paired version retains the direct t singleton, pools B, BH with base weights, and pools $q, q + 1$ without weights. The staged version pools t, B, BH with base weights and again uses the unweighted $q, q + 1$ pool. All final equations remain sums of squares of polynomial residuals.

4.2 A parity argument replaces the removed factorial certificate

For the moderate-coefficient source set

$$t = 128 + 256u, \qquad K = t^4,$$

and retain the request $F = t^K$. The smaller-source alternative sets

$$t = 16 + 32u, \qquad K = 2^{20}t.$$

All parameters here range over all integers. In both cases t is a nonzero multiple of four. In the first case K is positive and even; in the second, the signed-power theorem forces $K \geq 0$ and thus $t > 0$. In either case the first power gives a positive even F with $F > |t|$.

The retained CRT row reads

$$Bv = C + F + 1, \qquad C = 2c.$$

Its right-hand side is odd. Consequently B is odd and nonzero, independently of its eventual binomial interpretation. Also $H = 2 + 4S$ for an integer S, so $|BH| \geq 2$. Its exact power gives

$$x \geq 0, \qquad q = (BH)^x.$$

Hence $q \neq 0, -1$, and all five bases are nonzero. This proves that every one of the old power requests is recovered by the new pools. It also justifies the unweighted two-base variants, because the nonzero consecutive integers q and $q + 1$ have the same sign.

4.3 The endpoint removes the remaining negative horizon

For the moderate-coefficient source, the sign of t has not yet been recovered. The retained beta endpoint rows give

$$y = F(2v + 1), \qquad g = 1 + (t + 1)y, \qquad x = 1 + gu_t.$$

Since $|t| \geq 128$ and $|y| \geq F$,

$$|g| \geq |t + 1|F - 1 \geq 127F - 1 > 1.$$

Thus $x = 0$ would force $gu_t = -1$, a contradiction. It follows that $x \geq 1$ and $|q| \geq 2$. The literal retained request is q^t, with nominal index t; its signed exactness forces $t \geq 0$. Therefore $t \geq 128$.

This argument uses neither a missing factorial row nor a false conclusion from the binomial high quotient. In particular, setting $q = 1$ in the digit row does not identify B alone: the high quotient remains. The endpoint factorization is the required noncircular exclusion of $x = 0$.

4.4 Completeness and literal polynomial projections

The positive branches of both affine progressions are cofinal and divisible by four. Package 173's completeness construction can choose its horizon in either progression. It needs precisely these properties and $K \geq 2^{20}t$, rather than the earlier square representation. The CRT anchor C can still be chosen positive, even, and larger than every other nominal index. All subsequent cap and arithmetic witnesses can be completed.

Conversely the preceding sign deductions restore the hypotheses of the retained arithmetic decoding theorem. The same label-separation and Horn-flow arguments yield a finite C222 refutation, and the identified inherited C222/ZF/ZFC calibration gives a finite ZFC refutation. Thus the new sources have a zero exactly when the fixed ZFC presentation has a finite refutation.

Every subsequent projection uses a residual $\epsilon v + f = 0$ with $\epsilon \in \{1, -1\}$ and v absent from f. The supplied definition lists are acyclic. Substitution therefore gives a bijection on integer zero sets, with the

integral polynomial reconstruction $v = -\epsilon f$. Registers in the final circuits are abbreviations and do not add quantified variables.

The retained profiles are as follows. The last column bounds literal circuit constants, not fully expanded polynomial coefficients.

Variables	Exact degree	Residuals	Operations	Literal constant bound
131	12	70	2475	1099511627780
132	12	71	2484	256
117	16	56	2439	256
108	32	47	2434	256
97	128	36	2418	256
91	256	30	2400	256
87	32768	26	2388	256
86	2097780	25	2385	256

The immediate Oriented Powers comparisons include 137 variables at degree 12, 121 at degree 16, 111 at degree 32, 99 at degree 128, and 87 at degree 8388612. These are matched construction improvements, not claims of global minimality. Exact degrees are certified by nonzero leading homogeneous specializations; propagated coefficient-norm estimates provide separate explicit upper bounds on expanded coefficient height.

The source checker independently binds all sixteen actual requests, fourteen canonical Pell pairs, their shifted multiplication wires and norms, the complete orientation row, and every squared frame atom. The circuit checker verifies local polynomial expansions, acyclic unit pivots, source-row coverage, degree certificates, and a nonzero derivative witness for each quantified variable. These finite checks authenticate the construction; they do not replace its unbounded arithmetic proof or the inherited logical calibration.

5 Why the consistency calibration is an ordinary ZFC proof

The direct and compact branches have different immediate calibrations. The direct branch encodes the numerical proof calculus C222. It does not simulate K215, instantiate Jones's universal polynomial, or use Sun's integer-domain reduction. The compact branch deliberately retains its existing native/K215 coefficient construction. The new argument there is an exact soundness restoration to P66 and a fresh-witness completeness construction; it does not replace the inherited coefficient calibration by mere numerical inequalities.

5.1 Direct branch

The square-shell arithmetic constructors are

$$I(a,b) = 2((a+b)^2 + a), \qquad M(a,b) = I(a,b) + 1,$$
$$A(a,b) = 2((a+b)^2 + 3a + 4b + 3), \qquad E(a,b) = A(a,b) + 1.$$

They are the actual constructors in the arithmetic source. Their constructor-preserving recoding to the original numerical C222 syntax fixes truth and falsity. One must not identify the two sets of numeric constructors literally or assume that recoding preserves arithmetic on their labels.

The inherited endpoints used after that recoding are

`KEndpoint.c222_iff_zf_refutation`,

`ConstructibleChoice.zf_refutation_iff_zfc_refutation`.

The pinned source files are included for inspection and cited directly [5, 6]. They contain the named theorem compositions. Their full transitive import closure is not rebuilt by this submission.

The lower C222 direction uses first-order completeness, named formulas and a seed-to-model construction; the upper direction uses an interpretation in a constructible model. The ZF/ZFC transport uses the constructible submodel together with soundness and completeness. These are ordinary constructions available in ZFC. The internally well-founded part of the lower model is an *internal* notion; no assumption of an externally well-founded or transitive set model is added.

The arithmetic reductions are uniform over finite syntax, integers and finite lists. Their inductions, finite CRT choices, Pell recurrences, finite polynomial identities and finite group arguments are formalizable in ZFC. Composing them with the identified logical endpoints gives the internal equivalence in Theorem 1.1, without assuming consistency. The entire inherited calibration is not described as a newly implemented primitive-recursive proof translator: its supplied proof includes semantic completeness and model constructions.

5.2 Compact branch

The four fixed coefficient ports are bound to one complete Start-only4127 native construction. Their full bytes, a finite reconstruction recipe and a reconstructing program are supplied. Their total raw size is 97,046 bytes, and the largest fixed integer has 194,440 bits. The new residue-ordinate proof reconstructs all missing positive P66 witnesses from a new zero. Conversely a refutation supplies the canonical outer P66 witnesses, after which new private strong and auxiliary Pell witnesses are chosen. This distinction matters: arbitrary old auxiliary witnesses need not satisfy the new residue condition.

No coefficient port from another encoding is substituted. The inequalities $\beta \geq 31$, $0 < M_C < \beta < M_F < 2\beta$ are useful in the proof, but they do not alone give a consistency calibration. That calibration belongs to the whole named, authenticated tuple.

5.3 Proof boundary

The new unbounded arguments have received separately assigned AI-assisted mathematical reviews. The checks supplied with the archive authenticate finite algebra and exact source binding. They do not replace the Pell classification theorem, the native window theorem, or the C222/ZF/ZFC calibration by testing examples.

No new complete Lean or Rocq proof term or encoded object-language ZFC derivation is supplied. This is an ordinary proof with exact computational certificates and explicit inherited dependencies. It should be integrated with that status rather than described as newly kernel-checked.

6 Verification, reproducibility, and integration

6.1 What the programs check

The quartic checker is independent of its emitter. It checks the exact source bytes, the protected-row edits, commutation with the inherited projection and optimization chain, the affine substitution, the two invertible row operations, every coefficient of the collected sums of squares, degree four and occurrence of every variable. The signed horizon is a semantic change; the checker does not mislabel it as a pointwise inverse to the old square horizon.

The compact checker reconstructs the literal graphs and their fixed coefficient binding. It verifies all 56 terms of each leading form and the 323-term main-norm cancellation. The integer-domain checker authenticates the eleven positivity substitutions, the three free integer coordinates, the 145-gate count, and the nonzero degree-288 leader. All four coefficient payloads are independently regenerated from the complete fixed recipe and compared byte for byte.

The pooled-source checker binds sixteen actual power requests and fourteen canonical Pell pairs, including every shifted multiplication wire, norm, orientation term and required squared frame atom.

The circuit checker verifies all unit pivots and source-row partitions, local polynomial expansions, degree witnesses, coefficient bounds and actual dependence on each input. Deliberate corruptions of relevant artifact fields are rejected. A fresh reconstruction reproduces its mathematical JSON files exactly.

The release commands run the substantive checks both normally and under `python -O`. Verification does not search for a zero and cannot decide consistency. Degree statements use exact leading forms or nonzero finite-field specializations of the highest homogeneous part, not floating-point sampling.

6.2 Files and commands

The root directory contains this article, a root verifier, integration notes, a SHA-256 manifest and three mathematical subpackages: `quartic/`, `compact/` and `direct/`. Each contains the relevant proof, literal equations and replay tools. The final mathematical definitions are the exact integer residual lists, straight-line circuits, fixed coefficient bytes and their stated domains. Named intermediate registers are polynomial abbreviations, not hidden quantified variables.

From the extracted package:

```
python3 -B verify_all.py
python3 -O -B verify_all.py
```

Python 3.10 or later and its standard library suffice. The report can be rebuilt with an ordinary TeX Live installation:

```
cd report
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

PDF metadata can vary with the TeX installation. Frozen-byte manifests and mathematical reconstruction have deliberately separate roles. The integration notes suggest an additive new package; existing calibrations and historical equations should remain available.

7 Further research questions

1. **A second residue in the same ordinate.** Can an auxiliary Pell ordinate authenticate another quotient already present in the compact formula without increasing the number of free witnesses or losing the common-sign unit argument? The present proof depends on $M \mid v$, $M \mid S^2$ and $R < M$; a second congruence would have to be compatible with the same cofinal witness construction.
2. **Smaller carrier modulus.** The q^4 carrier follows from the proven bound $R < q^4 - 2$. A smaller modulus would need a sharper bound on the recovered index difference, rather than an unsupported assertion that congruent indices are equal.
3. **Sharper constructor growth budgets.** The direct quartics use the uniform depth constant $\ell = 2^{20}$. For a nonnegative-coefficient polynomial $A(t)$ of degree at most e, the finite inequality $A(t_0) < t_0^e$ implies $A(t) < t^e$ for all $t \geq t_0$, by monotonicity of each term of $A(t)/t^e$. Applying this to the actual constructor DAG can improve its coarse depth budget. Extending a final horizon below eight still requires rechecking every CRT and digit estimate; no such additional equation is asserted here.
4. **Reduce quartic height after the sign change.** The balanced Q_{214} trades a variable for coefficient height. Further integral coordinate or residual changes may reduce its 655,360 height without restoring a horizon variable. Any claimed improvement should be measured on the fully collected polynomial.
5. **Simplify the logical source.** The direct arithmetic cost is still driven by the large guarded C222 schema and its constructor tree. A different equiconsistent numerical calculus might improve more than local certificate edits. Such a route must supply its own uniform calibration, including the converse direction.

6. **Close the formalization boundary.** The new sign recovery, residue-ordinate extraction, protected-row commutation and exact circuit certificates are natural formalization targets. Their finite algebra does not by itself close the inherited unbounded arithmetic and logical dependency chain.

A Complete fixed coefficient recipe

The following finite recipe is retained from the whole Start-only4127 construction [2]. Its fixed indices do not introduce quantified variables into any equation. The accompanying four unsigned big-endian payloads have a combined size of 97,046 bytes.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{47}, \quad B = \rho^{4127}, \quad \beta = B - 1,$$
$$G_m(x) = \sum_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{2042} + \rho^{3074}.$$

$$\begin{aligned} M_C &= \beta + \rho - (\rho+3)\rho^{990} \\ &\quad - G_{943}(\rho) - \Psi(\rho, 1), \\ M_F &= B + 3 + (2^{44} - 2)\Lambda \\ &\quad + \rho^{973} G_9(\rho) + \rho^{1012} G_8(\rho) + \rho^{1031} \\ &\quad + 2^{29}\rho^{4124} - \rho^{4106}, \\ K_0 &= \rho^{20} + \rho^{30} + \rho^{1031} + \rho^{4137} \\ &\quad + \Lambda\Phi(1/\rho), \end{aligned}$$

where

$$\begin{aligned} \Phi(x) &= \sum_{i=0}^{942} [1 + 2^{10}(1 + w_i + 2^8 n_i \\ &\quad + 2^{17} p_i + 2^{25} s_i)] x^i + 2^{10}\Psi(x, 2), \\ \Psi(x, t) &= x^{992}(x^{10} + t^{17}) G_8(tx) \\ &\quad + t^8 x^{943}(x^{30} + t^{17}) G_9(tx) \\ &\quad + t^{34}(2 - t) x^{1031}. \end{aligned}$$

The rational intermediate in K_0 cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$h(q, b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\},$$
$$r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \ldots, q-1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j+2\ (j = 2q+b < 430)$	$((1-d_j)r_{0,u_j}, h(q,b), d_j r_{1,u_j}, v_j)$
$434 - 2a + b\ (a, b = 0, 1)$	$(0, 432a+b, 0, 432a+b)$
$\substack{424+2r_{d,q}+b \\ q\in\mathcal{S}_d,\ d,b=0,1}$	$(dr_{d,q}, b, (1-d)r_{d,q}, h(q,b))$
$924 + 2a + b\ (a, b = 0, 1)$	$(0, 432a+b, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t\ (t = 434, 435, 436)$	$(1, 438, 1, t)$
$934 + 3a + b\ (a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A,k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430+q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14 + m_q(D_{16}(E_0, \tau_q) - 14),$$
$$u_j = \nu_j(1+\sigma_j) + (1-\nu_j)D_{216}(T_0, j-\sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b+1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is:

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

B Complete literal balanced quartic

Every $x_0, \ldots, x_{213}$ below ranges independently over $\mathbb{Z}$. The equation is

$$\boxed{Q_{214}(x_0, \ldots, x_{213}) = \sum_{j=0}^{152} R_j^2 = 0.}$$

The list is generated directly from the verified integer coefficient file. It contains 153 quadratic residuals; there are no unstated side conditions or auxiliary quantifiers. The adjacent JSON also records every collected coefficient of the one quartic.

$$\begin{aligned}
R_0 &= 8 - 9x_3 - 16x_3x_{184} + x_6 + 16x_{184} \\
R_1 &= -x_1 - x_3x_4 + x_5x_6 \\
R_2 &= -1 + x_0 - x_5x_6 \\
R_3 &= -2x_1x_{212} - x_2x_{17} + x_3 + 2x_{15} \\
R_4 &= -2 - x_7 - x_7^2 - 2x_7x_8 - 2x_8 \\
&\quad - x_8^2 + x_{18} \\
R_5 &= -2 - x_7 - x_7^2 - 8x_7x_{18} - 4x_7x_{19} \\
&\quad - 8x_{18} - 16x_{18}^2 - 16x_{18}x_{19} - 4x_{19} - 4x_{19}^2 \\
&\quad + x_{20} \\
R_6 &= 1 + 2x_{20} - 4x_{20}^2 + x_{21} \\
R_7 &= -2 - x_8^2 - 2x_8x_9 - x_9^2 + x_{22} \\
R_8 &= -1 + 2x_7 - x_7^2 - 2x_7x_9 + 2x_9 \\
&\quad - x_9^2 + x_{23} \\
R_9 &= -7 + 33x_7 - 35x_7^2 - 96x_7x_9 - 24x_7x_{18} \\
&\quad - 4x_7x_{22} - 24x_7x_{23} + 16x_8 - 2x_8^2 - 8x_8x_{22} \\
&\quad + 32x_9 - 64x_9^2 - 32x_9x_{18} - 32x_9x_{23} + 4x_{18} \\
&\quad - 4x_{18}^2 - 8x_{18}x_{23} + 12x_{22} - 4x_{22}^2 + 8x_{23} \\
&\quad - 4x_{23}^2 + x_{24} \\
R_{10} &= -16 + 14x_7 - 2x_7^2 - 8x_7x_8 - 8x_7x_{22} \\
&\quad + 8x_8 + 8x_{22} - 4x_{24}^2 + x_{25} - 2x_{42} \\
R_{11} &= -6 + 16x_9 - 16x_9^2 - 16x_9x_{22} - 16x_9x_{23} \\
&\quad + 8x_{22} - 4x_{22}^2 - 8x_{22}x_{23} + 8x_{23} - 4x_{23}^2 \\
&\quad + x_{26} \\
R_{12} &= -5 + 24x_9 - 16x_9^2 - 16x_9x_{18} - 32x_9x_{23} \\
&\quad - 16x_9x_{26} + 10x_{18} - 4x_{18}^2 - 16x_{18}x_{23} - 8x_{18}x_{26} \\
&\quad + 24x_{23} - 16x_{23}^2 - 16x_{23}x_{26} + 12x_{26} - 4x_{26}^2 \\
&\quad + x_{27} \\
R_{13} &= -12 - 29x_7 - 37x_7^2 - 24x_7x_8 - 112x_7x_9 \\
&\quad - 4x_7x_{22} - 24x_7x_{23} - 39x_8 - 37x_8^2 - 112x_8x_9 \\
&\quad - 24x_8x_{22} - 4x_8x_{23} - 80x_9 - 128x_9^2 - 32x_9x_{22} \\
&\quad - 32x_9x_{23} - 12x_{22} - 4x_{22}^2 - 8x_{23} - 4x_{23}^2 \\
&\quad + x_{28} \\
R_{14} &= -18 - 10x_7 - 2x_7^2 - 24x_7x_8 - 32x_7x_9 \\
&\quad - 8x_7x_{22} - 72x_8 - 72x_8^2 - 192x_8x_9 - 48x_8x_{22} \\
&\quad - 96x_9 - 128x_9^2 - 64x_9x_{22} - 24x_{22} - 8x_{22}^2 \\
&\quad - 4x_{28}^2 + x_{29}
\end{aligned}$$

$$
\begin{aligned}
R_{15} &= -32 - 42x_7 + 8x_7x_9 + 16x_7x_{22} - 84x_8 \\
&\quad + 16x_8x_9 + 32x_8x_{22} + 21x_9 - 3x_9^2 - 4x_9x_{18} \\
&\quad - 8x_9x_{22} + 21x_{18} - 8x_{18}x_{22} + 24x_{22} - 4x_{22}^2 \\
&\quad + x_{30} - x_{58} \\
R_{16} &= -1 - 24x_7 - 64x_7^2 - 160x_7x_9 - 64x_7x_{22} \\
&\quad - 32x_7x_{23} - 32x_7x_{30} - 26x_9 - 100x_9^2 - 80x_9x_{22} \\
&\quad - 40x_9x_{23} - 40x_9x_{30} - 8x_{22} - 16x_{22}^2 - 16x_{22}x_{23} \\
&\quad - 16x_{22}x_{30} - 6x_{23} - 4x_{23}^2 - 8x_{23}x_{30} - 4x_{30} \\
&\quad - 4x_{30}^2 + x_{31} \\
R_{17} &= 1 - 4x_{18}^2 + x_{19} \\
R_{18} &= -2 - x_7 - x_7^2 + 8x_7x_{18} - 4x_7x_{19} \\
&\quad + 8x_{18} - 16x_{18}^2 + 16x_{18}x_{19} - 4x_{19} - 4x_{19}^2 \\
&\quad + x_{32} \\
R_{19} &= 1 + 2x_{32} - 4x_{32}^2 + x_{33} \\
R_{20} &= -x_8^2 - 4x_8x_{33} - 4x_{33}^2 + x_{34} \\
R_{21} &= -8 - 20x_7 - 16x_7^2 - 64x_7x_8 + 32x_7x_{18} \\
&\quad - 40x_8 - 64x_8^2 + 64x_8x_{18} + 18x_{18} - 16x_{18}^2 \\
&\quad + x_{35} \\
R_{22} &= 8x_8 - 62x_8^2 - 92x_8x_9 - 32x_8x_{22} - 32x_8x_{35} \\
&\quad + 6x_9 - 34x_9^2 - 24x_9x_{22} - 24x_9x_{35} - 4x_{22}^2 \\
&\quad - 8x_{22}x_{35} + 4x_{35} - 4x_{35}^2 + x_{36} \\
R_{23} &= 8x_7 - 64x_7^2 - 96x_7x_9 - 32x_7x_{23} - 32x_7x_{36} \\
&\quad + 6x_9 - 36x_9^2 - 24x_9x_{23} - 24x_9x_{36} + 2x_{23} \\
&\quad - 4x_{23}^2 - 8x_{23}x_{36} + 4x_{36} - 4x_{36}^2 + x_{37} \\
R_{24} &= -x_9^2 - 2x_9x_{10} - x_{10}^2 + x_{38} \\
R_{25} &= -32 + 22x_7 - 3x_7^2 - 8x_7x_8 - 8x_7x_{22} \\
&\quad - 8x_7x_{23} - 4x_7x_{38} + 16x_8 - 8x_8x_{23} - 2x_9^2 \\
&\quad - 8x_9x_{38} + 16x_{22} - 8x_{22}x_{23} + 22x_{23} - 4x_{23}^2 \\
&\quad + 4x_{38} - 4x_{38}^2 + x_{39} - x_{42} \\
R_{26} &= 6 - 16x_7^2 - 36x_7x_9 - 32x_7x_{23} - 40x_7x_{38} \\
&\quad - 16x_7x_{39} - 22x_9^2 - 32x_9x_{23} - 48x_9x_{38} - 16x_9x_{39} \\
&\quad - 2x_{23} - 16x_{23}^2 - 32x_{23}x_{38} - 16x_{23}x_{39} + 4x_{38} \\
&\quad - 24x_{38}^2 - 16x_{38}x_{39} - 4x_{39}^2 + x_{40} \\
R_{27} &= x_9 - 25x_9^2 - 40x_9x_{23} - 20x_9x_{26} - 16x_{23}^2 \\
&\quad - 16x_{23}x_{26} - 4x_{26}^2 + x_{41} \\
R_{28} &= -1 + 8x_8 - 4x_8^2 - 8x_8x_{22} + 8x_{22} \\
&\quad - 4x_{22}^2 + x_{42} \\
R_{29} &= -1 - 16x_8^2 - 8x_8x_9 - 32x_8x_{22} - 16x_8x_{42} \\
&\quad - x_9^2 - 8x_9x_{22} - 4x_9x_{42} - 16x_{22}^2 - 16x_{22}x_{42} \\
&\quad - 4x_{42}^2 + x_{43} \\
R_{30} &= 3 + x_8 - 2x_8x_9 - 4x_8x_{43} - 3x_9^2 \\
&\quad - 8x_9x_{43} - x_{22} - 4x_{43}^2 + x_{44} \\
R_{31} &= -1 - 4x_{44} - 4x_{44}^2 + x_{45} \\
R_{32} &= -1 - x_9^2 - 8x_9x_{44} - 4x_9x_{45} - 16x_{44}^2 \\
&\quad - 16x_{44}x_{45} - 4x_{45}^2 + x_{46} \\
R_{33} &= 2 + x_8 - x_8^2 - 4x_8x_9 - 4x_8x_{46}
\end{aligned}
$$

$$
\begin{aligned}
&\quad - 4x_9^2 - 8x_9x_{46} - 4x_{46}^2 + x_{47} \\
R_{34} &= -4x_{41}^2 - 8x_{41}x_{47} - 4x_{47}^2 + x_{48} \\
R_{35} &= -9 + 7x_7 - x_7^2 - 8x_7x_{41} - 4x_7x_{48} \\
&\quad + 24x_{41} - 16x_{41}^2 - 16x_{41}x_{48} + 12x_{48} - 4x_{48}^2 \\
&\quad + x_{49} \\
R_{36} &= -1 - 4x_{49} - 4x_{49}^2 + x_{50} \\
R_{37} &= -x_9^2 - 8x_9x_{49} - 4x_9x_{50} - 16x_{49}^2 - 16x_{49}x_{50} \\
&\quad - 4x_{50}^2 + x_{51} \\
R_{38} &= -x_7 - x_7^2 - 4x_7x_9 - 4x_7x_{51} - 4x_9 \\
&\quad - 4x_9^2 - 8x_9x_{51} - 4x_{51} - 4x_{51}^2 + x_{52} \\
R_{39} &= -10 - 12x_{52} - 4x_{52}^2 + x_{53} \\
R_{40} &= -19 + 4x_8 - 4x_8^2 - 8x_8x_{22} - 16x_8x_{52} \\
&\quad - 8x_8x_{53} + 4x_{22} - 4x_{22}^2 - 16x_{22}x_{52} - 8x_{22}x_{53} \\
&\quad - 16x_{52} - 24x_{52}^2 - 16x_{52}x_{53} + 6x_{53} - 4x_{53}^2 \\
&\quad + x_{54} \\
R_{41} &= -10 + x_8 - 25x_8^2 - 40x_8x_{22} - 20x_8x_{54} \\
&\quad - 16x_{22}^2 - 16x_{22}x_{54} - 12x_{47} - 4x_{47}^2 - 4x_{54}^2 \\
&\quad + x_{55} \\
R_{42} &= 21 + 24x_{47} - 8x_{47}^2 - 16x_{47}x_{55} - 2x_{55} \\
&\quad - 4x_{55}^2 + x_{56} \\
R_{43} &= -5 + 4x_7 + 8x_8 + x_9 - x_9^2 \\
&\quad - 4x_9x_{56} - 2x_{18} - 4x_{56}^2 + x_{57} + 12x_{58} \\
&\quad - 4x_{58}^2 \\
R_{44} &= -6x_7 - 16x_7^2 - 64x_7x_8 + 8x_7x_9 + 16x_7x_{18} \\
&\quad - 12x_8 - 64x_8^2 + 16x_8x_9 + 32x_8x_{18} + 3x_9 \\
&\quad - x_9^2 - 4x_9x_{18} + 3x_{18} - 4x_{18}^2 + x_{58} \\
R_{45} &= -12 + 8x_7 + 16x_8 - 4x_{18} - 4x_{57}^2 \\
&\quad + 24x_{58} - 8x_{58}^2 + x_{59} \\
R_{46} &= -1 - 4x_{59} - 4x_{59}^2 + x_{60} \\
R_{47} &= -x_1x_{12} + x_{61} \\
R_{48} &= x_{11} - x_{12} - 2x_{21}x_{61} - x_{62} \\
R_{49} &= x_{11} - x_{12} - 2x_{25}x_{61} - x_{63} \\
R_{50} &= x_{11} - x_{12} - 2x_{27}x_{61} - 6x_{61} - x_{64} \\
R_{51} &= x_{11} - x_{12} - 2x_{29}x_{61} - x_{65} \\
R_{52} &= x_{11} - x_{12} - 2x_{31}x_{61} + 2x_{61} - x_{66} \\
R_{53} &= x_{11} - x_{12} - 4x_{33}x_{61} - 2x_{34}x_{61} - x_{67} \\
R_{54} &= x_{11} - x_{12} - 2x_{37}x_{61} - x_{68} \\
R_{55} &= 2x_{27}x_{61} - 2x_{40}x_{61} - 6x_{61} + x_{64} - x_{69} \\
R_{56} &= -x_7x_8 + x_7x_9 + x_8 + x_8x_9 - x_9 \\
&\quad - x_9^2 + x_{70} \\
R_{57} &= 1 + 2x_{10} + x_{10}^2 - x_{70}^2 + x_{71} \\
R_{58} &= -x_{11}^2 + 2x_{11}x_{12} + 2x_{11}x_{73} - x_{12}^2 - 2x_{12}x_{73} \\
&\quad - x_{71}^2 + x_{72} - x_{73}^2 \\
R_{59} &= -4x_{59}x_{61} - 2x_{60}x_{61} + x_{73} \\
R_{60} &= -x_8x_{61} - x_{12} + x_{74}x_{75} + x_{76} \\
R_{61} &= -1 + 4x_1 + 4x_1x_7 + 8x_1x_8 - 2x_1x_{18} \\
&\quad + x_{74}
\end{aligned}
$$

$$
\begin{aligned}
R_{62} &= -x_7x_{207} - x_{11} + x_{75} + x_{207}\\
R_{63} &= x_8x_{207} - 2x_{18}x_{61} + 2x_{31}x_{61} + x_{66} + x_{77}\\
R_{64} &= -x_{11}x_{62} + x_{12}x_{62} + x_{78}\\
R_{65} &= -x_{63}x_{78} + x_{79}\\
R_{66} &= -x_{64}x_{79} + x_{80}\\
R_{67} &= -x_{65}x_{80} + x_{81}\\
R_{68} &= -x_{66}x_{81} + x_{82}\\
R_{69} &= -x_{67}x_{82} + x_{83}\\
R_{70} &= -x_{68}x_{83} + x_{84}\\
R_{71} &= -x_{69}x_{84} + x_{85}\\
R_{72} &= -x_{76}x_{85} + x_{86}\\
R_{73} &= -x_{77}x_{86} + x_{87}\\
R_{74} &= -x_{87}^2 + x_{88}\\
R_{75} &= -x_3 - 2x_3x_{15} + 2x_{15} + x_{89}\\
R_{76} &= -x_{13}x_{89} + x_{90}\\
R_{77} &= -2x_0^2 + 2x_0x_{11} + 2x_0x_{12} + 2x_0x_{90} + 2x_0x_{91}\\
&\quad - x_{11}^2 - 2x_{11}x_{90} - x_{12}^2 - 2x_{12}x_{91} + x_{16}x_{93}\\
&\quad - x_{72}x_{88} - x_{90}^2 - x_{91}^2\\
R_{78} &= -x_3x_{14} + x_{14} - x_{14}x_{89} + x_{91}\\
R_{79} &= -x_{94}x_{97} + x_{97} + x_{99}\\
R_{80} &= -x_2x_{120} + x_{95}x_{199} - x_{98} - x_{99}x_{120}\\
R_{81} &= x_{100}x_{199} - x_{101}x_{123} - x_{102}\\
R_{82} &= x_{103}x_{199} - x_{104}x_{123} - x_{105}\\
R_{83} &= x_{106}x_{199} - x_{107}x_{123} - x_{108}\\
R_{84} &= x_{109}x_{199} - x_{110}x_{123} - x_{111}\\
R_{85} &= x_{112}x_{200} - x_{114}x_{123} - x_{115}\\
R_{86} &= x_{116}x_{200} - x_{117}x_{123} - x_{118}\\
R_{87} &= -x_{16}x_{120} + x_{123}\\
R_{88} &= x_{119} - x_{121}x_{123} - x_{122}\\
R_{89} &= -x_{123}x_{125} + x_{124} - x_{126}\\
R_{90} &= -x_{123}x_{128} + x_{127}x_{199} - x_{129}\\
R_{91} &= x_{130} - 2x_{130}x_{132} + x_{133}\\
R_{92} &= -x_{130}x_{133} - x_{133} + x_{134}\\
R_{93} &= -1 + x_{131}^2 + x_{133}^2 - x_{134}^2\\
R_{94} &= 1 - x_{131}x_{136} - x_{135} + x_{137}\\
R_{95} &= -1 + 2x_{135} - x_{135}^2 + 2x_{135}x_{137} + x_{136}^2\\
R_{96} &= 65 - 16x_{131} - 32x_{131}x_{184} + x_{138} + 256x_{184}\\
&\quad + 256x_{184}^2\\
R_{97} &= x_1 + x_{138}x_{139} - x_{180} - 8x_{181} - 16x_{181}x_{184}\\
R_{98} &= 1 + x_2^2 - 2x_2x_{131} + x_{140}\\
R_{99} &= -x_2x_{172} + x_{16} + x_{140}x_{141} - x_{171}\\
R_{100} &= -1 + 2x_{131}x_{183} - x_{142} - x_{183}^2\\
R_{101} &= -1 + x_{94} + x_{142}x_{143} - x_{171} - x_{172}x_{183}\\
R_{102} &= -x_{131}x_{145} - x_{144} + x_{146}\\
R_{103} &= -1 - x_{144}^2 + 2x_{144}x_{146} + x_{145}^2\\
R_{104} &= 1 - x_{131}x_{148} - x_{147} + x_{149}\\
R_{105} &= -1 + 2x_{147} - x_{147}^2 + 2x_{147}x_{149} + x_{148}^2\\
R_{106} &= -x_{131}x_{151} - x_{150} + x_{152}\\
R_{107} &= -1 - x_{150}^2 + 2x_{150}x_{152} + x_{151}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{108} &= -x_{131}x_{154} - x_{153} + x_{155} \\
R_{109} &= -1 - x_{153}^2 + 2x_{153}x_{155} + x_{154}^2 \\
R_{110} &= -x_{131}x_{157} - x_{156} + x_{158} \\
R_{111} &= -1 - x_{156}^2 + 2x_{156}x_{158} + x_{157}^2 \\
R_{112} &= -x_{131}x_{160} - x_{159} + x_{161} \\
R_{113} &= -1 - x_{159}^2 + 2x_{159}x_{161} + x_{160}^2 \\
R_{114} &= 1 - x_{131}x_{163} - x_{162} + x_{164} \\
R_{115} &= -1 + 2x_{162} - x_{162}^2 + 2x_{162}x_{164} + x_{163}^2 \\
R_{116} &= -x_{131}x_{166} - x_{165} + x_{167} \\
R_{117} &= -1 - x_{165}^2 + 2x_{165}x_{167} + x_{166}^2 \\
R_{118} &= -x_{131}x_{169} - x_{168} + x_{170} \\
R_{119} &= -1 - x_{168}^2 + 2x_{168}x_{170} + x_{169}^2 \\
R_{120} &= 1 - x_{131}x_{172} - x_{171} + x_{173} \\
R_{121} &= -1 + 2x_{171} - x_{171}^2 + 2x_{171}x_{173} + x_{172}^2 \\
R_{122} &= -x_{131}x_{175} - x_{174} + x_{176} \\
R_{123} &= -1 - x_{174}^2 + 2x_{174}x_{176} + x_{175}^2 \\
R_{124} &= -x_{131}x_{178} - x_{177} + x_{179} \\
R_{125} &= -1 - x_{177}^2 + 2x_{177}x_{179} + x_{178}^2 \\
R_{126} &= 1 - x_{131}x_{181} - x_{180} + x_{182} \\
R_{127} &= -1 + 2x_{180} - x_{180}^2 + 2x_{180}x_{182} + x_{181}^2 \\
R_{128} &= -21 - x_0^2 - 16x_1 - 2x_1^2 - 32x_1x_{184} \\
&\quad - x_2^2 - x_3^2 + 2x_7 - x_7^2 - x_8^2 \\
&\quad - x_9^2 - x_{10}^2 - x_{11}^2 - x_{12}^2 - x_{13}^2 \\
&\quad - x_{14}^2 - 4x_{15}^2 - x_{16}^2 - x_{92}^2 + 2x_{94} \\
&\quad - 2x_{94}^2 - x_{95}^2 - x_{96}^2 - x_{100}^2 - x_{103}^2 \\
&\quad - x_{106}^2 - x_{109}^2 - x_{112}^2 - x_{113}^2 - x_{116}^2 \\
&\quad - x_{119}^2 - x_{120}^2 - x_{124}^2 - x_{127}^2 + x_{130} \\
&\quad - x_{183}^2 - 32x_{184} - 2x_{213} \\
R_{129} &= 2x_2 - 4x_2x_7 - 4x_2x_8 - 4x_2x_9 - 4x_2x_{10} \\
&\quad - 4x_2x_{11} - 4x_2x_{12} - 4x_2x_{13} - 4x_2x_{14} - 8x_2x_{15} \\
&\quad + x_{183} \\
R_{130} &= -x_{131}x_{186} + x_{195} \\
R_{131} &= -1 + x_{185}^2 + x_{186}^2 - x_{195}^2 \\
R_{132} &= -x_{188}x_{189} + x_{196} \\
R_{133} &= -1 + x_{187}^2 + x_{188}^2 - x_{196}^2 \\
R_{134} &= -x_{178}^2 + x_{197} \\
R_{135} &= x_{186} - 2x_{191}x_{197} - x_{197} \\
R_{136} &= -1 - 4x_{186} - 8x_{186}x_{190} + x_{189} \\
R_{137} &= -x_{131} - x_{185}x_{192} + x_{189} \\
R_{138} &= -x_{179} - x_{185}x_{193} + x_{187} \\
R_{139} &= -2x_{15} - 4x_{178}x_{194} + x_{188} \\
R_{140} &= 9 + x_0^2 - 2x_0x_{172} + 16x_1 + x_1^2 \\
&\quad - 2x_1x_{145} + 32x_1x_{184} + x_3^2 - 2x_3x_{163} - 2x_7 \\
&\quad + x_7^2 - 2x_7x_{148} + x_8^2 - 2x_8x_{151} + x_9^2 \\
&\quad - 2x_9x_{154} + x_{10}^2 - 2x_{10}x_{157} + x_{11}^2 - 2x_{11}x_{160} \\
&\quad + x_{12}^2 - 2x_{12}x_{166} + x_{13}^2 - 2x_{13}x_{169} + x_{14}^2 \\
&\quad - 2x_{14}x_{175} - 4x_{15}x_{178} + x_{92}^2 - 2x_{92}x_{181} - x_{131}x_{198}
\end{aligned}
$$

$$
\begin{aligned}
&- 16x_{136} - 32x_{136}x_{184} + x_{137}^2 - 16x_{145} - 32x_{145}x_{184} \\
&- 2x_{146} + x_{146}^2 + 2x_{148} + x_{149}^2 - 2x_{152} \\
&+ x_{152}^2 - 2x_{155} + x_{155}^2 - 2x_{158} + x_{158}^2 \\
&- 2x_{161} + x_{161}^2 + x_{164}^2 - 2x_{167} + x_{167}^2 \\
&- 2x_{170} + x_{170}^2 + x_{173}^2 - 2x_{176} + x_{176}^2 \\
&+ 2x_{178}^2 - 2x_{179} + x_{179}^2 + x_{182}^2 + x_{198} \\
&- x_{201}^2 - x_{202}^2 - x_{203} - x_{203}^2 + x_{213} \\
R_{141} &= -x_{96}x_{199} + x_{120} \\
R_{142} &= -x_{113}x_{200} + x_{120} \\
R_{143} &= -1 - 4x_{15}^2 - x_{136}^2 - x_{145}^2 - x_{148}^2 \\
&- x_{151}^2 - x_{154}^2 - x_{157}^2 - x_{160}^2 - x_{163}^2 \\
&- x_{166}^2 - x_{169}^2 - x_{172}^2 - x_{175}^2 + x_{178}^2 \\
&- x_{181}^2 - x_{201}^2 - x_{202}^2 - x_{203} - x_{203}^2 \\
R_{144} &= -1 - 4x_{98}^2 - 4x_{102}^2 - 4x_{105}^2 - 4x_{108}^2 \\
&- 4x_{111}^2 - 4x_{115}^2 - 4x_{118}^2 + x_{120}^2 - 4x_{122}^2 \\
&- 4x_{126}^2 - 4x_{129}^2 - x_{204}^2 - x_{205}^2 - x_{206} \\
&- x_{206}^2 \\
R_{145} &= -x_1x_{11} + x_{207} \\
R_{146} &= x_{94}x_{209} - 2x_{95}x_{145} - 2x_{100}x_{148} - 2x_{103}x_{151} - 2x_{106}x_{154} \\
&- 2x_{109}x_{157} - 2x_{112}x_{160} - 2x_{116}x_{166} - 2x_{119}x_{169} - 2x_{124}x_{175} \\
&- 2x_{127}x_{178} - 2x_{131}x_{209} + 2x_{145}x_{146} + 2x_{148} + 2x_{148}x_{149} \\
&+ 2x_{151}x_{152} + 2x_{154}x_{155} + 2x_{157}x_{158} + 2x_{160}x_{161} + 2x_{166}x_{167} \\
&+ 2x_{169}x_{170} + 2x_{175}x_{176} + 2x_{178}x_{179} + x_{208} \\
R_{147} &= -10 - x_{94}x_{208} + x_{95}^2 - 2x_{95}x_{144} + x_{100}^2 \\
&- 2x_{100}x_{147} + x_{103}^2 - 2x_{103}x_{150} + x_{106}^2 - 2x_{106}x_{153} \\
&+ x_{109}^2 - 2x_{109}x_{156} + x_{112}^2 - 2x_{112}x_{159} + x_{116}^2 \\
&- 2x_{116}x_{165} + x_{119}^2 - 2x_{119}x_{168} + x_{124}^2 - 2x_{124}x_{174} \\
&+ x_{127}^2 - 2x_{127}x_{177} + 2x_{144}x_{146} + 2x_{147} + 2x_{147}x_{149} \\
&+ 2x_{150}x_{152} + 2x_{153}x_{155} + 2x_{156}x_{158} + 2x_{159}x_{161} + 2x_{165}x_{167} \\
&+ 2x_{168}x_{170} + 2x_{174}x_{176} + 2x_{177}x_{179} + x_{209} \\
R_{148} &= x_{94}x_{211} - 2x_{96}x_{136} - 2x_{113}x_{163} - 2x_{120}x_{172} - 2x_{131}x_{211} \\
&+ 2x_{136} + 2x_{136}x_{137} + 2x_{163} + 2x_{163}x_{164} + 2x_{172} \\
&+ 2x_{172}x_{173} + x_{210} - x_{211} \\
R_{149} &= -2 - x_{94}x_{210} + x_{96}^2 - 2x_{96}x_{135} + x_{113}^2 \\
&- 2x_{113}x_{162} + x_{120}^2 - 2x_{120}x_{171} + 2x_{135} + 2x_{135}x_{137} \\
&+ 2x_{162} + 2x_{162}x_{164} + x_{171}^2 - x_{172}^2 + x_{210} \\
&+ x_{211} \\
R_{150} &= -1 - x_1 - 2x_1x_{212} + x_3 \\
R_{151} &= -128 - 512x_{184} - 512x_{184}^2 + x_{213} \\
R_{152} &= x_{92} - 512x_{213}^2
\end{aligned}
$$

C Source identities and comparison notes

The exact repository source identities, original Git blob identifiers and SHA-256 digests are retained in each subpackage's provenance ledger. The earlier Oriented Powers archive has SHA-256

`b5f564d8afc22cd0d17fd831f965baf9fa3423155229c7caa9b1084cddcbbaa2`.

The final archive's manifest authenticates its delivered contents. Inherited notices identify prior authorship and source scope. The new manuscript and code are AI-assisted work prepared for ProveIt; separate AI reviews are not described as external human peer review.

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[2] ProveIt contributors, *Start-marker compositions*, package 174, 25 October 2026. Complete Start-only4127 tuple and P66 calibration. https://github.com/VladimirReshetnikov/ProveIt/tree/02a78a08603cbae248c19c23a75907c6c6bd6443/Computability/DiophantineConsistency/con-zfc-diophantine-forms/174-start-marker-compositions.

[3] ProveIt contributors, *Quotient carriers*, package 170, 25 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/02a78a08603cbae248c19c23a75907c6c6bd6443/Computability/DiophantineConsistency/con-zfc-diophantine-forms/170-quotient-carriers.

[4] *Oriented power pooling and polynomial root centering*, AI-assisted research prepared for Vladimir Reshetnikov and ProveIt, 25 October 2026. Prior delivered archive `ConZFC_Oriented_Powers_2026-10-25.zip`; exact identity above. The local inherited theorem and source pins are in `direct/`.

[5] ProveIt contributors, `KEndpoint.lean`, `c222_iff_zf_refutation`. https://github.com/VladimirReshetnikov/ProveIt/blob/02a78a08603cbae248c19c23a75907c6c6bd6443/Computability/BusyBeaver/Lean/BusyBeaver/ZFC/KEndpoint.lean.

[6] ProveIt contributors, `ConstructibleChoice.lean`, `zf_refutation_iff_zfc_refutation`. https://github.com/VladimirReshetnikov/ProveIt/blob/02a78a08603cbae248c19c23a75907c6c6bd6443/Computability/BusyBeaver/Lean/BusyBeaver/ZFC/ConstructibleChoice.lean.

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[8] Zhi-Wei Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, 2 September 2026. https://arxiv.org/abs/2609.01594.